

Miaomiao Dai

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EDUCATION

University of Pennsylvania

M.S.E. in Robotics

Aug 2026 – present

Philadelphia, PA

University of Michigan

B.S.E. in Computer Science, **Summa Cum Laude** (SJTU-UMich dual-degree sequence) · GPA 3.8/4.0

Aug 2024 – May 2026

Ann Arbor, MI

- Coursework: Robot Learning (A+), Algorithmic Robotics (A), Machine Learning (A), Computer Vision (A)

Shanghai Jiao Tong University

B.E. in Mechanical Engineering, Global College · GPA 3.6/4.0

Aug 2022 – Aug 2024, May 2026 – Aug 2026

Shanghai, China

PUBLICATIONS

- [1] **Miaomiao Dai**, Zhongqiang Ren. “HOP: Fast Differential Dynamic Programming for Horizon-Optimal Trajectory Planning.” *Robotics: Science and Systems (RSS)*, 2026.
- [2] **Miaomiao Dai**, Zhongqiang Ren. “AL-HOP: Augmented-Lagrangian Horizon-Optimal Planning for Fast Constrained Trajectory Optimization.” In submission to *IEEE International Conference on Robotics and Automation (ICRA)*, 2027.
- [3] PhysLA: physical supervision for latent-action VLA pretraining (full title and author list withheld during double-blind review). Under review at *ICLR 2027*; **second author**.

RESEARCH EXPERIENCE

Measure the Value, Learn the Slope

Sep 2026 – present

Independent research, University of Pennsylvania · **first author**, in preparation for ICML 2027

- Proposed a **test-time correction for frozen world models**. When the world quietly changes (say, a motor 20% weaker), the model’s predicted landing point goes wrong, yet its slope (which way to move) still holds; so the method **measures where the action lands and takes only the direction from the model**, one small step per try, with nothing retrained.
- Pre-registered every test and re-ran it on fresh starts: on LeWM’s Reacher task with that weaker motor, success rises from **17% to 96%** in five tries; between the re-plans of a slow planner, **62% → 75%** on LeWM and **57% → 77%** on DINO-WM.

HOP: Horizon-Optimal Trajectory Planning

Jul 2025 – Feb 2026

Advisor: Prof. Zhongqiang Ren, Shanghai Jiao Tong University · **RSS 2026 talk, undergraduate first author**

- Proposed **HOP**, a new algorithm that makes LQR and DDP **horizon-optimal**: it optimizes the trajectory and its horizon together instead of fixing the horizon in advance.
- Proved the Riccati recursion is a **linear fractional transformation**, so all candidate horizons share one computation, $O(N^2) \rightarrow O(N)$ even when time-varying; HOP-DDP handles nonlinear systems, up to **40×** faster than brute force.
- Designed and ran the evaluation on four systems, from a double integrator to a 12-dimensional quadrotor.

AL-HOP: Constrained Horizon-Optimal Planning

Mar 2026 – Sep 2026

Advisor: Prof. Zhongqiang Ren, Shanghai Jiao Tong University · **first-author** manuscript in submission to ICRA 2027

- Proposed **AL-HOP**, extending HOP to hard state, control, and obstacle constraints: it alternates horizon selection on an augmented-Lagrangian LQ model with trajectory optimization, perturbing the horizon to escape local minima.
- Valid on **100%** of 500 runs (100 problems across 10 systems) vs. 79.8%, 79.8%, and 70% for ALTRO and direct collocation with IPOPT and SNOPT, and **2.4–4.8×** faster on problems both solved, at comparable cost.

PhysLA: Transferring Physical Supervision to Latent-Action VLA Pretraining

May 2026 – Sep 2026

Researcher, Shanghai Jiao Tong University · **second author**, under review at ICLR 2027

- Latent-action pretraining supervises contact and force only indirectly, and tactile or torque sensing covers only a small slice of video. PhysLA learns physical targets where sensors exist, infers them from causal RGB elsewhere, and uses them with visual prediction to supervise latent actions — **no tactile or torque sensing at deployment**.
- **Designed two of its three stages and co-implemented them**: learning **horizon-structured physical targets** from pressure, tactile, and torque measurements, and **transferring them to sensor-free video** via causal RGB.
- Outperforms VLA-JEPA, the strongest latent-action baseline tested, on LIBERO and every SimplerEnv task group, and beats both VLA-JEPA and $\pi_{0.5}$ on **each of five real-robot tasks**.

PROJECTS

A Voice-Interactive Closed-Loop VLA System for Low-Cost Tabletop Manipulation Feb 2026 – Aug 2026
B.E. thesis, Shanghai Jiao Tong University · **project lead**, team of 5 · Advisors: Prof. Hesheng Wang & Prof. Yunlong Guo

- **$\pi_{0.5}$ on real hardware**: collected **316 demonstrations** across 11 tabletop tasks, fine-tuned $\pi_{0.5}$ (LoRA on the 2B vision–language backbone, full 300M action expert), and **deployed it on a real SO-101 arm**.
- **Speech recognition**: integrated Qwen-Omni ASR with visual grounding to turn **spoken requests into task instructions**.
- **Toward closed-loop execution**: added visual checks of progress and task completion (terminal-verification F_1 **0.842**) with stop, re-observation, and bounded retries; the full system reached **70%** end-to-end success.

HONORS & AWARDS

James B. Angell Scholar, University of Michigan	Mar 2026
University Honors, University of Michigan (3 terms)	Dec 2024 – Dec 2025
Dean’s List, University of Michigan (3 terms)	Dec 2024 – Dec 2025
First Prize, 13th SJTU Liming Cup Mechanical Innovation Competition	May 2023
Honorable Mention, Interdisciplinary Contest in Modeling (ICM)	May 2023

SKILLS

Methods: trajectory optimization (DDP, augmented-Lagrangian methods), physical representation learning from pressure, tactile, and torque data, latent-action VLA pretraining, VLA fine-tuning & real-robot deployment ($\pi_{0.5}$), VLM integration
Languages: Python, C++ , C, MATLAB
ML & robotics: PyTorch, Transformers, OpenCV, NumPy, SciPy, OpenAI Gym, ROS, PyBullet, Gazebo
Tools: Git, Linux (Ubuntu), Docker, $\LaTeX$, SolidWorks